Skin Lesion Image Segmentation

Final Technical Summary

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1 Problem and Question

We segment skin lesions in dermoscopic images (binary mask: lesion vs. skin). **Question:** with the same pretrained encoder and the same training pipeline, does replacing the **U-Net** decoder by the **DeepLabV3+** decoder (atrous spatial pyramid pooling, larger receptive field) improve accuracy, and at what computational cost?

2 Dataset

ISIC 2018 Challenge, Task 1: 2,594 official training dermoscopic images with expert binary masks (206 different resolutions, from 576×540 to 6748×4499 px; the lesion covers 0.3–98.7% of the image, median 14%). Every file was checked against the official archive checksums. **Split:** ISIC contains repeated photographs of the same lesion, so a random split would leak copies into validation. We split by *groups*: images sharing a `patient_id` or `lesion_id` (ISIC metadata) or detected as near-duplicates (SIFT matching + RANSAC check on all 3.36 M image pairs, $\geq$8 inliers) always stay together: **2,205 train / 389 validation** images (85/15, seed 42). The official 1,000-image test set was not used. **Preprocessing:** resize to 256×256 (area interpolation; masks nearest-neighbour), ImageNet normalisation. Training augmentations: flips, 90° rotations, affine (scale 0.9–1.1, shift, ±30°), colour jitter.

3 Method and Experimental Setup

Baseline: U-Net; **second condition:** DeepLabV3+ (output stride 16, prediction at 1/4 resolution, one skip connection). Both use the same **ImageNet-pretrained ResNet-34** encoder (`segmentation_models_pytorch` 0.5.0), so only the decoder differs (24.4 M vs. 22.4 M parameters). Identical training: BCE + Dice loss, AdamW (lr 10^{-4}), batch 16, up to 50 epochs, early stopping on validation Dice (patience 10). Each model was trained with **three seeds** (42, 1, 2) to separate the architecture effect from run-to-run noise. **Metrics** (per image at threshold 0.5, then averaged): Dice, IoU, the ISIC thresholded Jaccard (IoU < 0.65 counts as 0), and HD95 boundary distance in pixels; plus training and inference time. **Ablation / sensitivity:** (i) the binarisation threshold (0.1–0.9) on the trained checkpoints (U-Net seed 42, DeepLabV3+ seeds 1, 2); (ii) the loss: BCE only and Dice only vs. BCE + Dice for U-Net, seed 42, everything else fixed.

4 Results

Validation, 389 images (mean ± std over 3 seeds)	**U-Net**	**DeepLabV3+**
Dice	**0.903 ± 0.002**	0.901 ± 0.002
IoU	**0.837 ± 0.003**	0.834 ± 0.003
ISIC thresholded Jaccard	**0.797 ± 0.002**	0.793 ± 0.005
Failed images (IoU < 0.65)	**32.3 ± 0.6**	33.3 ± 2.1
HD95 median / 95th percentile (px)	7.14 / 47.5	**6.87 / 43.1**
Dice, lesions < 5% of image ($n = 90$)	**0.883 ± 0.007**	0.875 ± 0.010
Inference per image (batch 1, MacBook Air M3, MPS)	**10.5 ms**	17.0 ms
Training time per epoch, same machine	**70 s**	94 s
Total training time (early stopping)	53.6 ± 12.4 min	**47.7 ± 2.3 min**

Loss ablation (U-Net, seed 42)	**Dice**	**IoU**	**ISIC Jaccard**	**Failed**	**HD95 p95**
BCE + Dice (baseline)	**0.901**	**0.834**	**0.796**	**32**	52.4
BCE only	0.897	0.828	0.778	41	**52.0**
Dice only	0.894	0.825	0.787	35	60.8

Table 1: Main comparison (top) and loss ablation (bottom); bold = better value. Per-seed numbers: `results/comparison/seeds_summary.md`; ablation: `results/ablation/`.

5 Interpretation

The two decoders segment practically equally well: U-Net is ahead by 0.0024 Dice on average and in each of the three seeds, but this gap is about the size of the run-to-run spread (std 0.002) and the 95% bootstrap interval of the per-image difference (−0.006 to +0.001) contains zero. The stable differences are small and follow from the architecture: DeepLabV3+ is worse on small lesions (−0.008 Dice for lesions under 5% of the image), because it predicts the mask at 1/4 resolution with only one skip connection, while its wider context gives fewer extreme border errors (HD95 95th percentile 43 vs. 48 px). DeepLabV3+ is also 1.3× slower to train per epoch and 1.6× slower at inference (its dilated last stage works on 16×16 maps). **Threshold:** 0.5 is near-optimal for both models (the best, 0.4, gains ≤0.0004 Dice; over 0.3–0.7 Dice changes by ≤0.006), so the comparison does not depend on it; only 0.1 or 0.9 hurt (−0.012 to −0.031). **Loss:** BCE + Dice is best on every overlap metric; dropping a term costs 0.004 (BCE only) or 0.007 (Dice only) Dice, but with one seed and bootstrap intervals that include zero this is a trend, not a significant effect. BCE only gives soft, poorly calibrated probabilities (Dice spread 0.027 over thresholds 0.3–0.7) and under-segments at 0.5 (9 more failures); Dice only gives near-binary outputs but stops early (best epoch 10) and has the worst boundary outliers (HD95 p95 60.8 vs. 52.4 px), because without the per-pixel BCE term isolated wrong regions are penalised less.

6 Failure Analysis

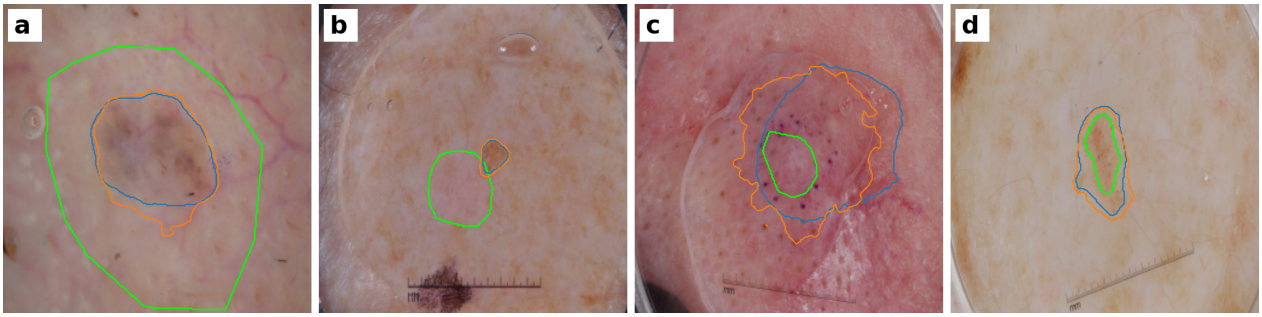


Figure 1: Failures shared by all six runs (IoU < 0.65 in every seed of both models). Green = ground truth, blue = U-Net, orange = DeepLabV3+.

13 images fail (IoU < 0.65) in all six runs, so these failures are properties of the image, not of one run. **(a) Annotation ambiguity** (Dice 0.38–0.44): the ground truth is a coarse polygon that includes a wide, barely coloured halo, while both models outline the visible pigmented centre. **(b) Wrong object** (Dice $\leq$0.04 in 5 of 6 runs): the annotated lesion is a faint pink, almost unpigmented melanoma; both models segment a nearby brown spot with higher contrast, because training data are dominated by brown, well-pigmented lesions. **(c) Artefact edge** (Dice 0.13–0.33): a fold of immersion fluid under the contact plate forms the strongest closed contour around a light lesion, and the models follow it. **(d) Small lesion with a diffuse border** (Dice 0.61–0.67 U-Net, 0.53–0.58 DeepLabV3+): both models over-segment, DeepLabV3+ more so in every seed, which matches its coarser output. Cases (b) and (c) also explain why the better boundary statistics of DeepLabV3+ do not reduce the number of failures: its wider context shortens border errors but does not help when the wrong region is chosen.

7 Conclusions and Limitations

With a shared pretrained encoder and pipeline, the decoder choice has almost no effect on overall accuracy (Dice $\approx$0.90 for both); the remaining errors come mostly from the data (annotation style, weakly pigmented lesions, contact-plate artefacts) and are shared by both models. The ablation supports the combined BCE + Dice loss and the default threshold 0.5 are sound choices. In our setting the simpler and faster **U-Net is the better choice**. **Limitations:** (i) all numbers are on our 389-image validation split, which was also used for early stopping, so they may be slightly optimistic; the official test set was not evaluated; (ii) three seeds, one split and one image size (256×256, which removes fine border detail); (iii) failure causes are read from the images, not tested by targeted experiments; (iv) timings come from fanless laptops that throttle (10–20% variation).

8 Reproducibility

Code: `https://github.com/amiranabiullina/skin_lesions_CV_project` (README with setup, data download, training, evaluation and ablation commands; committed split, configs, per-image metrics, logs and figures). **Environment:** Python 3.14, PyTorch 2.14.1 (MPS), segmentation_models_pytorch 0.5.0, Albumentations 2.0.8; pinned `requirements.txt`. Trained on Apple-silicon MacBook Air laptops (M3/M4, MPS): about 50 min per run; inference 10–17 ms per image.

9 Members' Contributions

Member	Contribution
Ilmira Usmanova	Evaluation script (Dice, IoU, HD95), result tables and figures
Ilia Kulichenko	Training script and U-Net baseline experiments
Alsu Nizamieva	DeepLabV3+ experiments and runtime measurements
Amira Nabiullina	Dataset code, preprocessing and splits; ablation; coordination, README and summary
Andrei Silantev	Boundary and failure-case analysis